디지털 논리

Karnaugh Map (I)

이 상 화

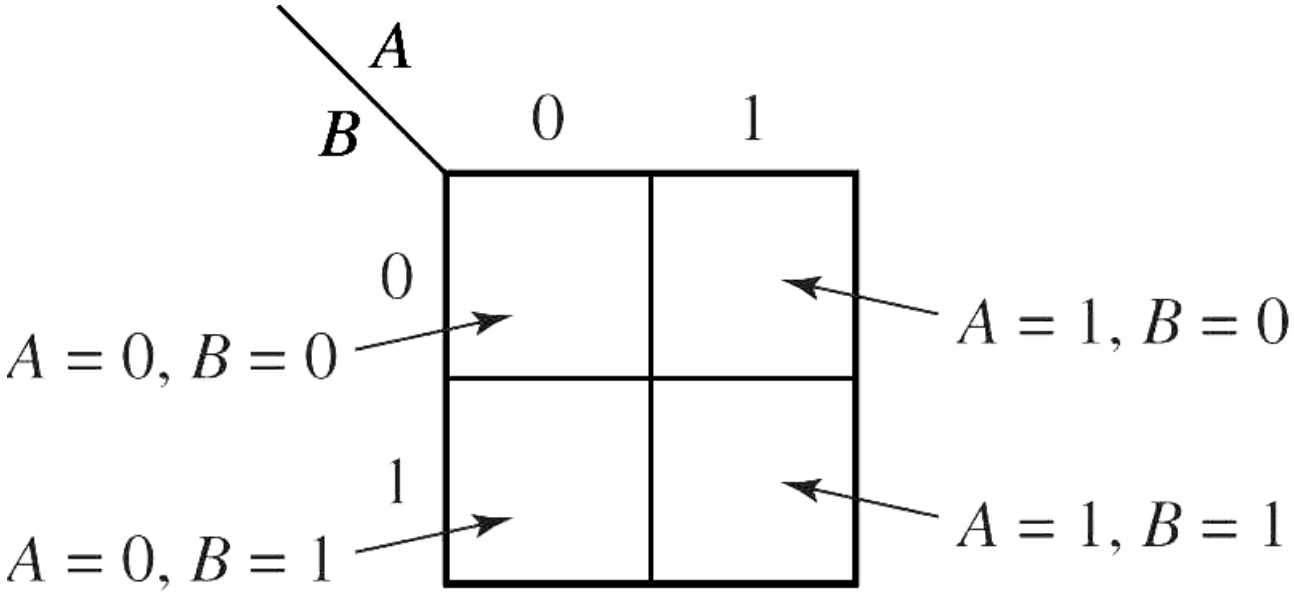
Karnaugh Map

- **Graphical approach for simplifying Boolean expressions with 2~ 5 variables**
  - Boolean algebra is difficult to simplify the expressions to the minimum forms.
- **Graphical representation of Boolean expressions to find minimum two-level circuits.**
  - Sum of product form (AND-OR)
- **On the meaning of “simplification”**
  - with minimum # of terms => minimum gates
  - With the smallest # of literals => minimum inputs

2-Variable Karnaugh Map (1)

F= AB’ + AB = A

F= A’B’ + A’B = A’



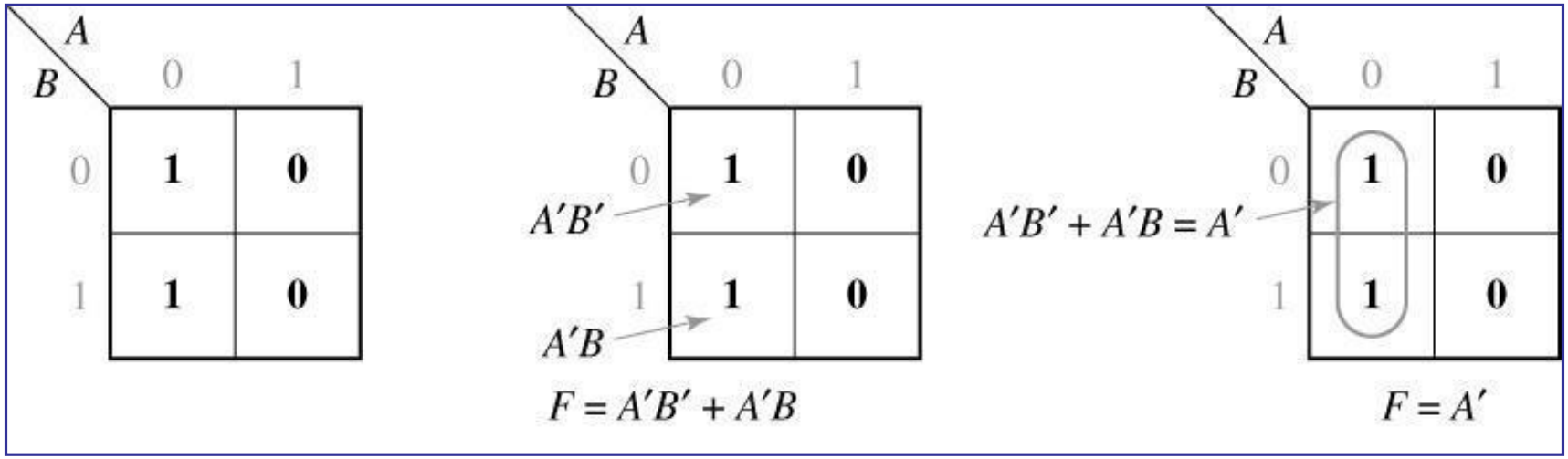
2-Variable Karnaugh Map (2)

Only one variable is different in the neighboring partition.
=> XY+XY'=X.

Ex:

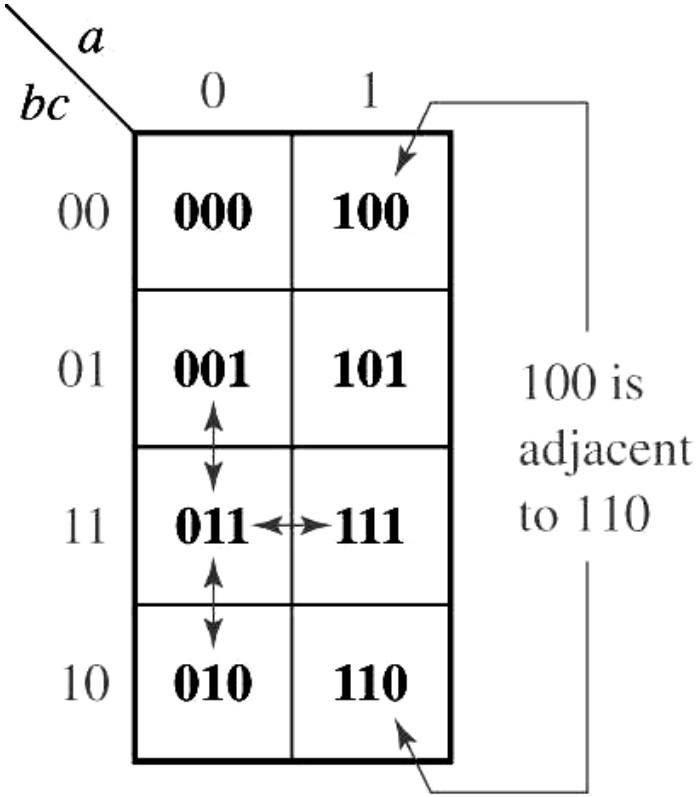
A	B	F
0	0	1
0	1	1
1	0	0
1	1	0

F= A'B' + A'B = A'



3-Variable Karnaugh Map (1)

$$F = A'BC + ABC = BC$$



(a) Binary notation

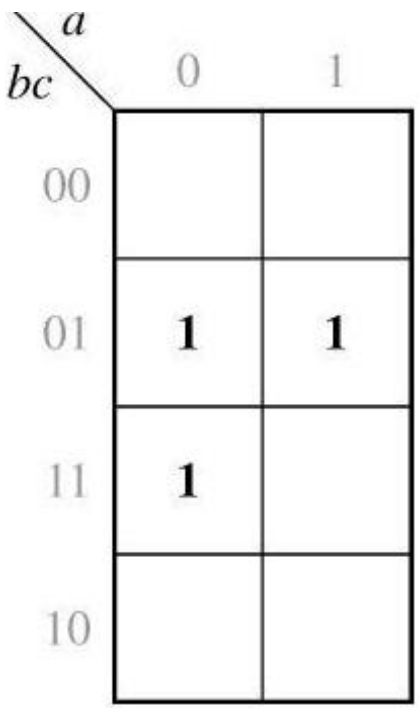
bc \ a	0	1
00	0	4
01	1	5
11	3	7
10	2	6

(b) Decimal notation

3-Variable Karnaugh Map (2)

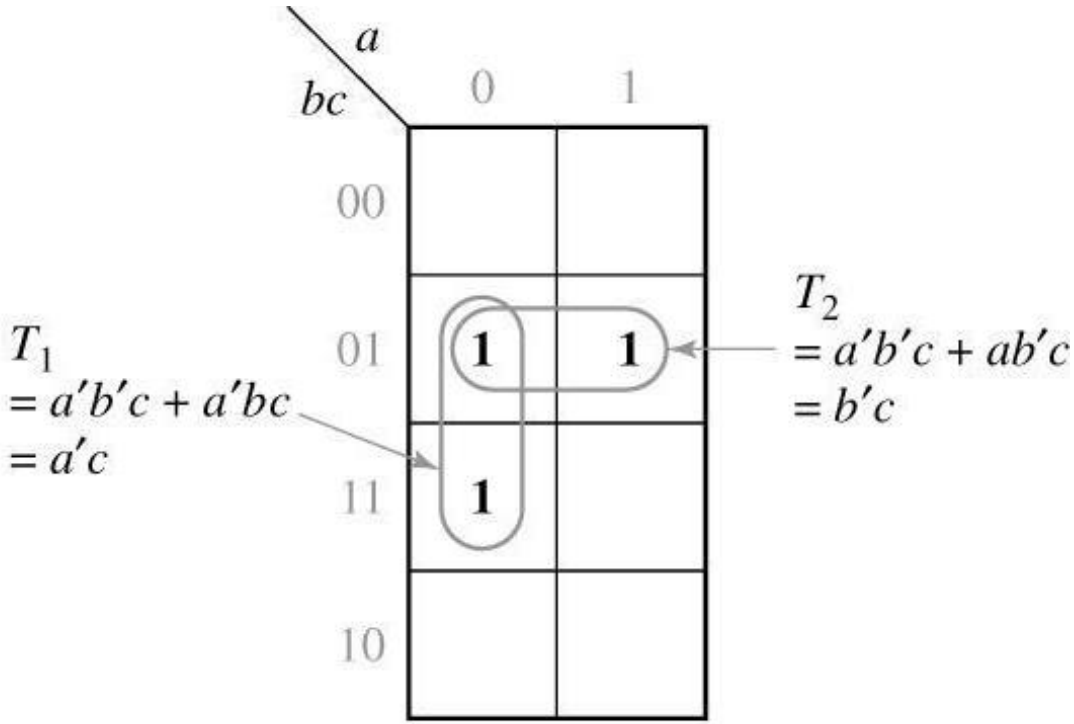
$$XY + XY' = X$$

Example



$F = \Sigma m\ (1, 3, 5)$

(a) Plot of minterms

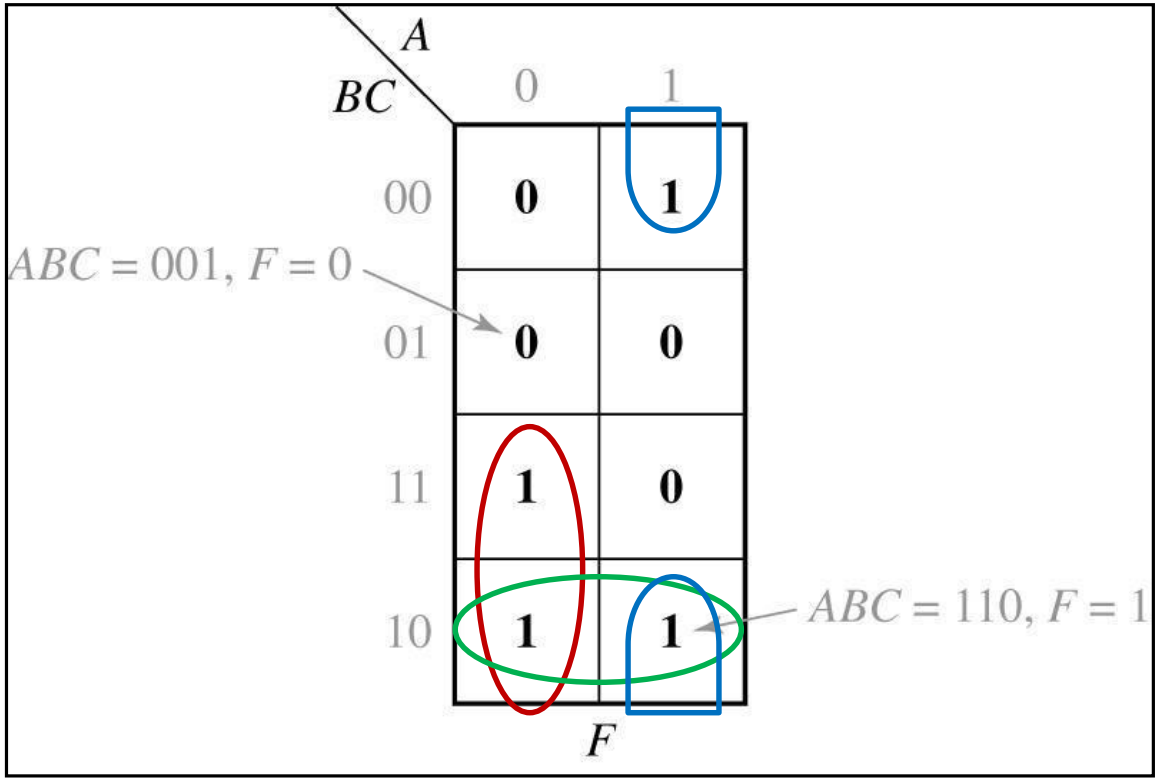


$$F = T_1 + T_2 = a'c + b'c$$

3-Variable Karnaugh Map (2)

Ex: Truth table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



A'B + BC' + AC'

By consensus theorem

Karnaugh Map of a Function

Given **minterm expansion**, the locations of corresponding minterms are assigned by **1 in Karnaugh map**.

Given *maxterm expansion*, the locations of corresponding maxterms are assigned by **0 in Karnaugh map**.

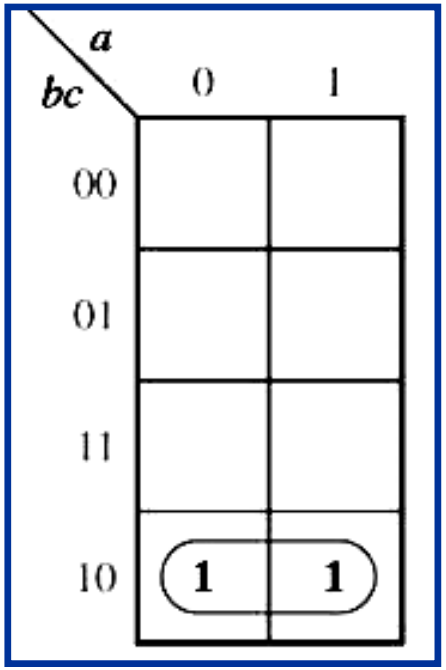
Ex: F(*a*, *b*, *c*) = ∑m(1, 3, 5) = ∏(0, 2, 4, 6, 7)

bc \ *a*	0	1
00	**0** 0	**0** 4
01	**1** 1	**1** 5
11	**1** 3	**0** 7
10	**0** 2	**0** 6

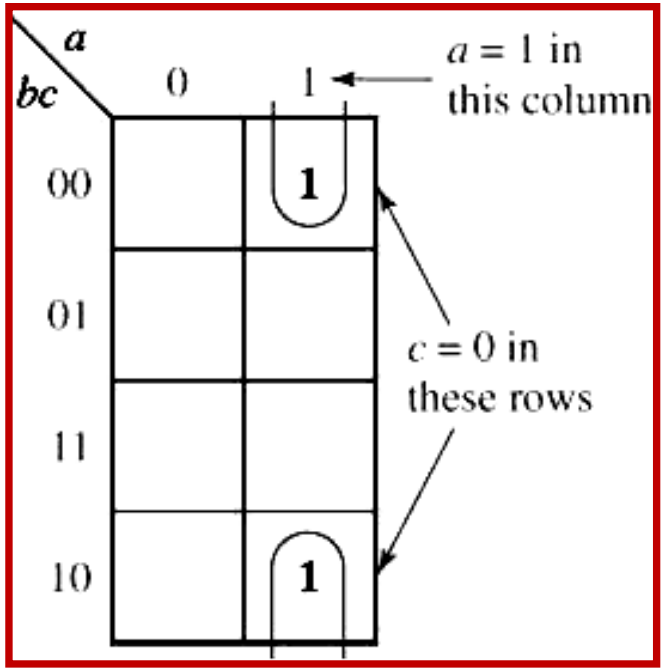
Karnaugh Map for Product Terms

Given a function algebraically, we assign 0 and 1 to the corresponding locations of product terms

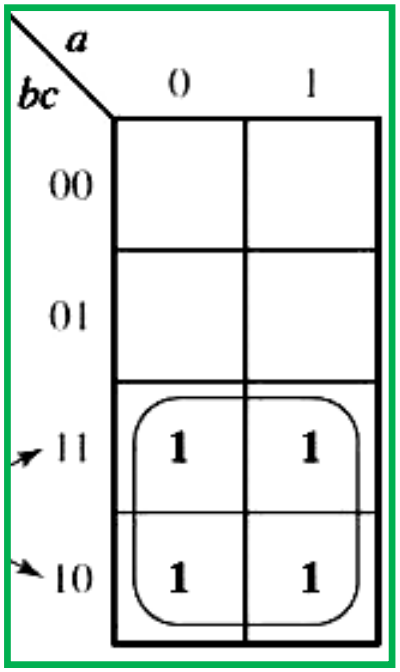
bc’



ac’



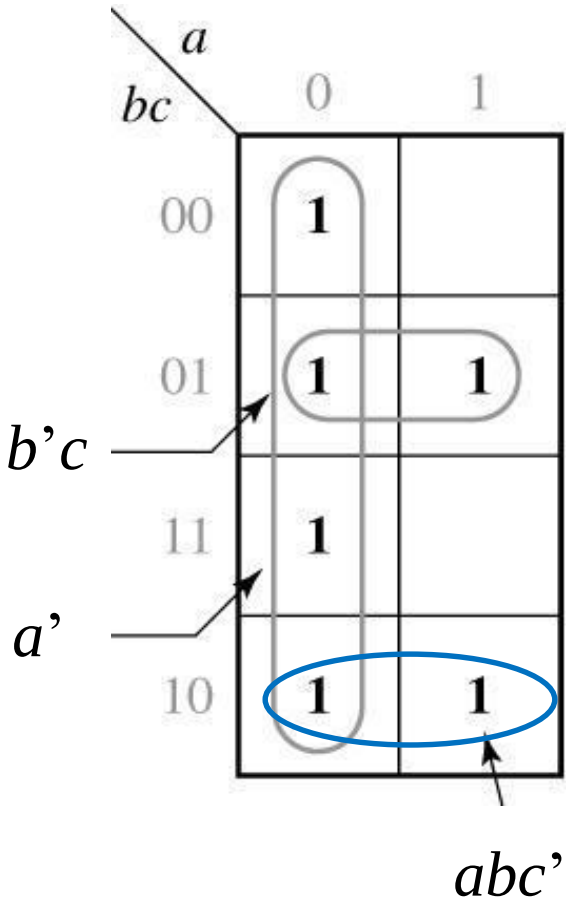
b



Karnaugh map for a given function

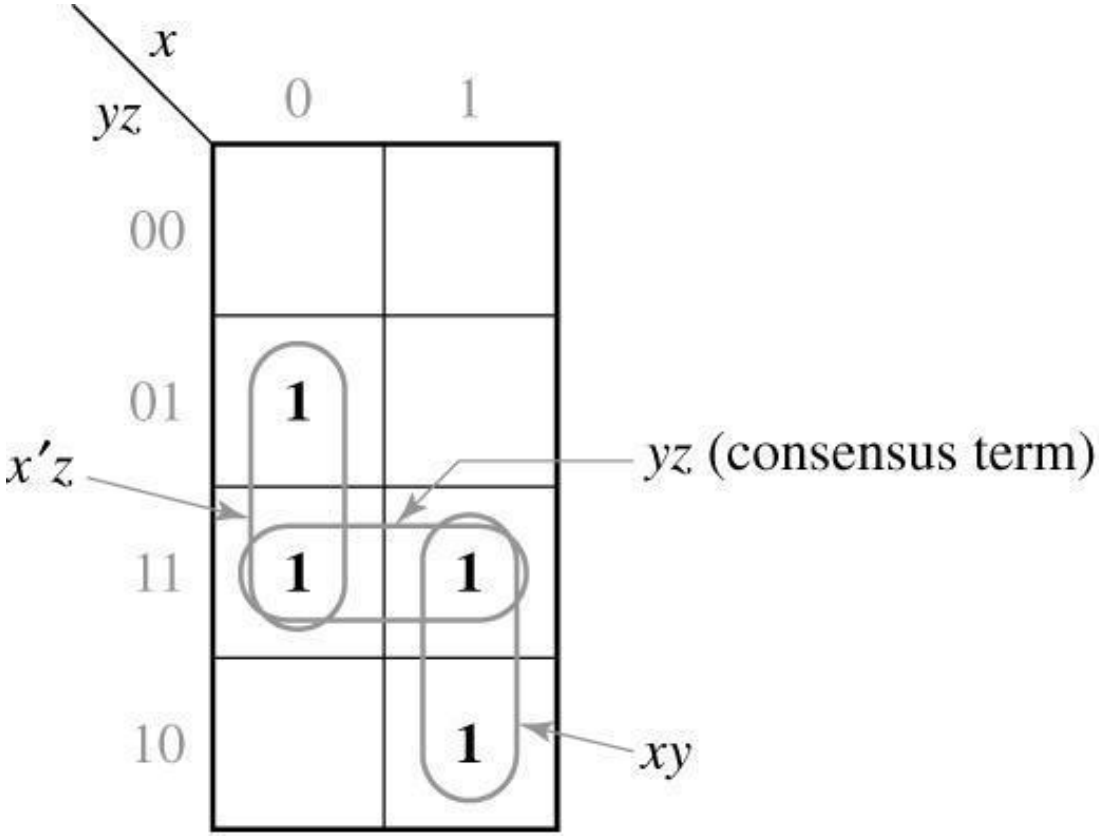
Given Function

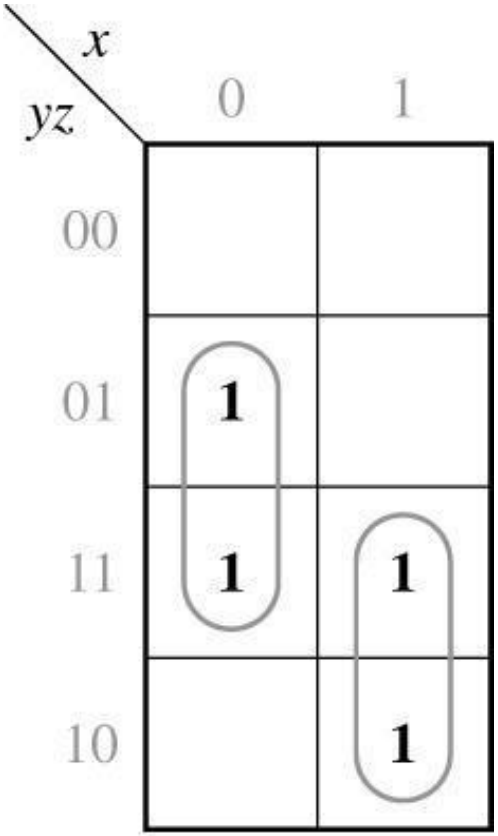
$$f(a,b,c) = abc' + b'c + a'$$



Consensus Theorem with K.map

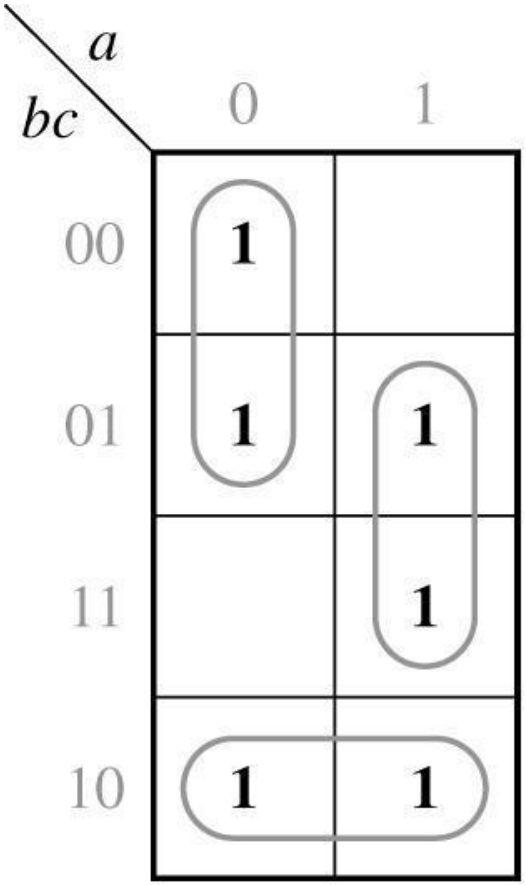
$$xy + x'z + yz = xy + x'y$$



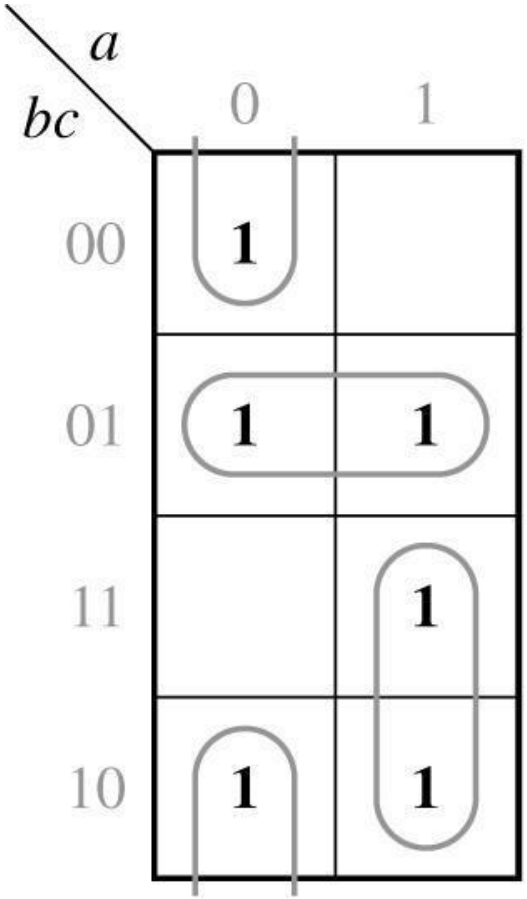


Function with 2 Minimum Forms

$$F = \sum m(0,1,2,5,6,7)$$



$F = a'b' + bc' + ac$



$F = a'c' + b'c + ab$

디지털 논리

Karnaugh Map (II)

이 상 화

Principles of Karnaugh Map

- **Grouping the adjacent 2/4/8… minterms**
  - Grouping should be rectangular
  - Every group of 2 minterms reduces one variable
  - Every group of 4 minterms reduces two variables
  - Every group of 8 minterms reduces three variable
- **Maximizing a group to cover all the adjacent minterms**
  - Adjacent 2 groups of 2 minterms => 1 group of 4 minterms
- **Minimizing the overlapped minterms when grouping**
  - Overlapping is necessary, but not too many.
  - Overlapping is the case of consensus theorem.

4-Variable Karnaugh Map

All the adjacent rows and columns have only one change of variable.

CD \ AB	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

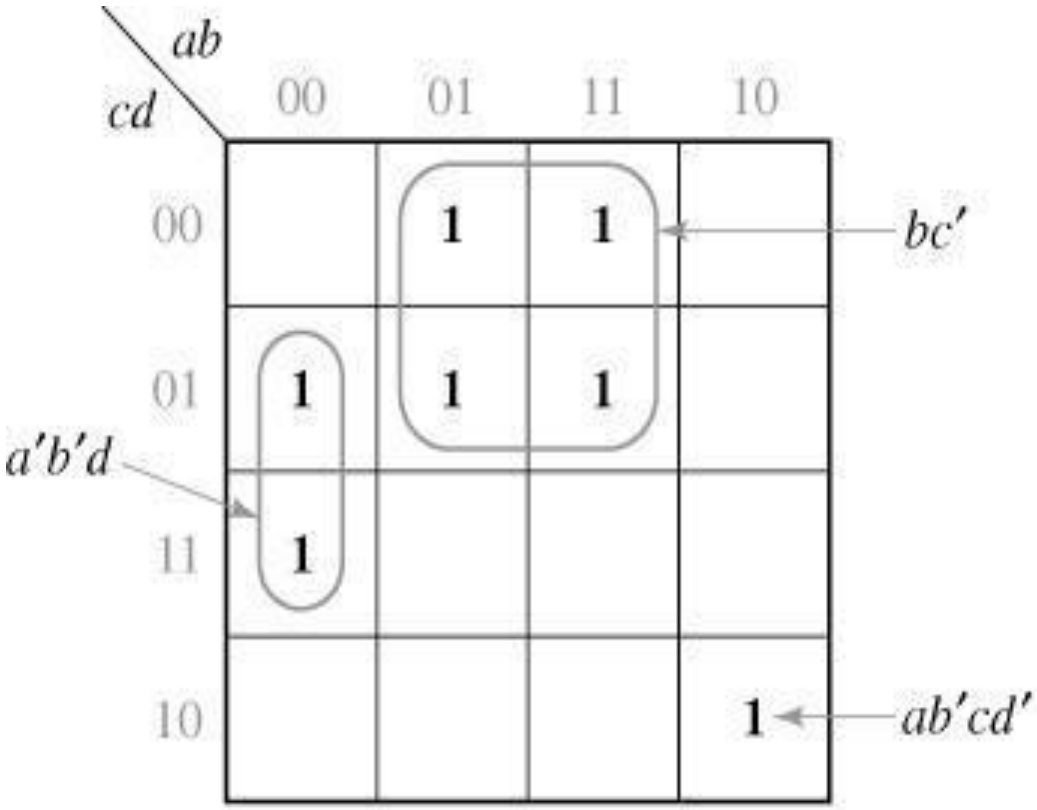
adjacent

adjacent

Ex: 4-Variable Karnaugh Map (1)

Given minterm expansion, use Karnaugh map to simplify the circuits

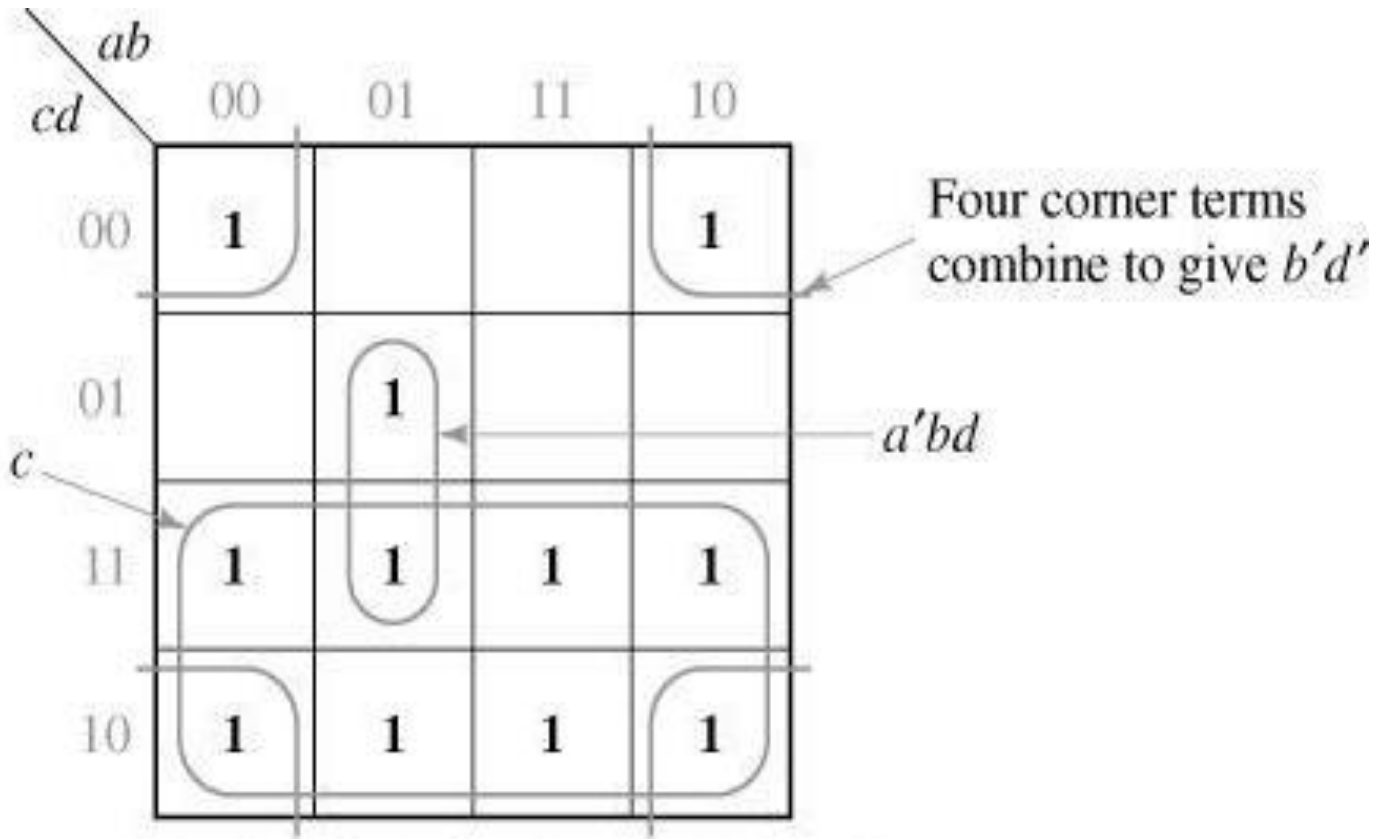
$$f_1 = \sum m(1,3,4,5,10,12,13) = bc' + a'b'd + ab'cd'$$



Ex: 4-Variable Karnaugh Map (2)

Given minterm expansion, use Karnaugh map to simplify the circuits

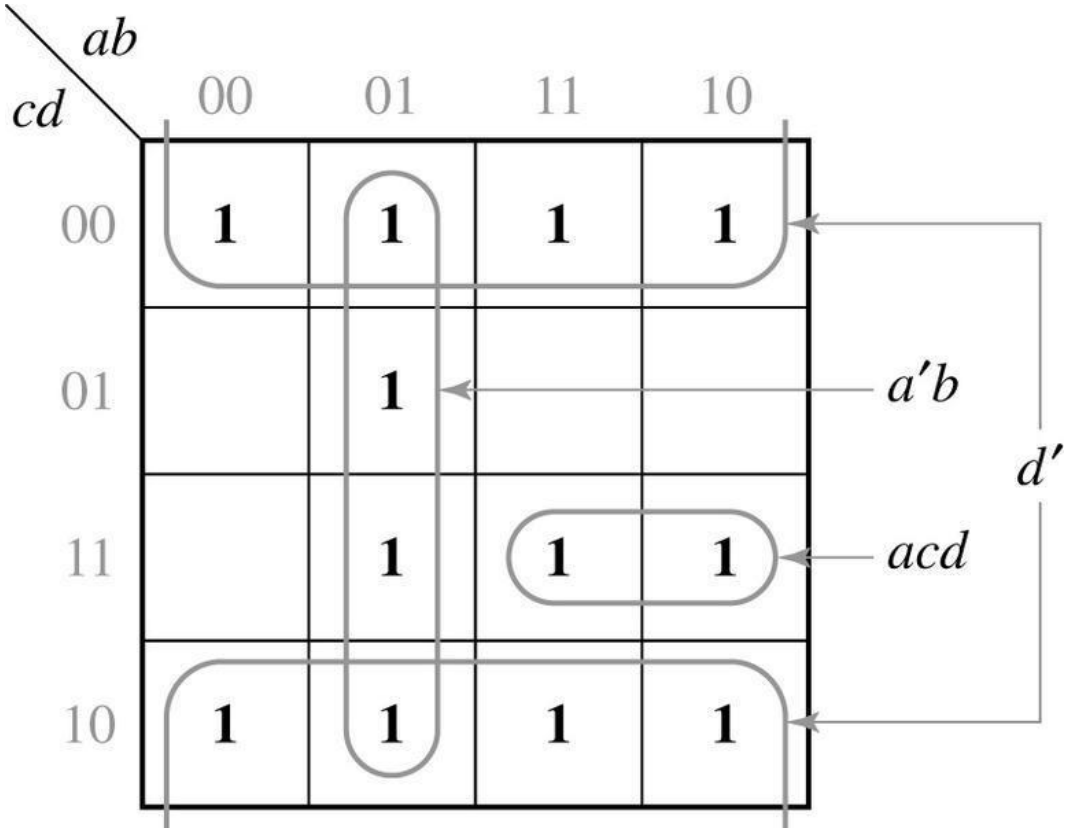
$$f_1 = \sum m(0,2,3,5,6,7,8,10,11,14,15) = c + b'd' + a'bd$$



Ex: 4-Variable Karnaugh Map (2)

Given expression of circuit, show the minterms on the Karnaugh map

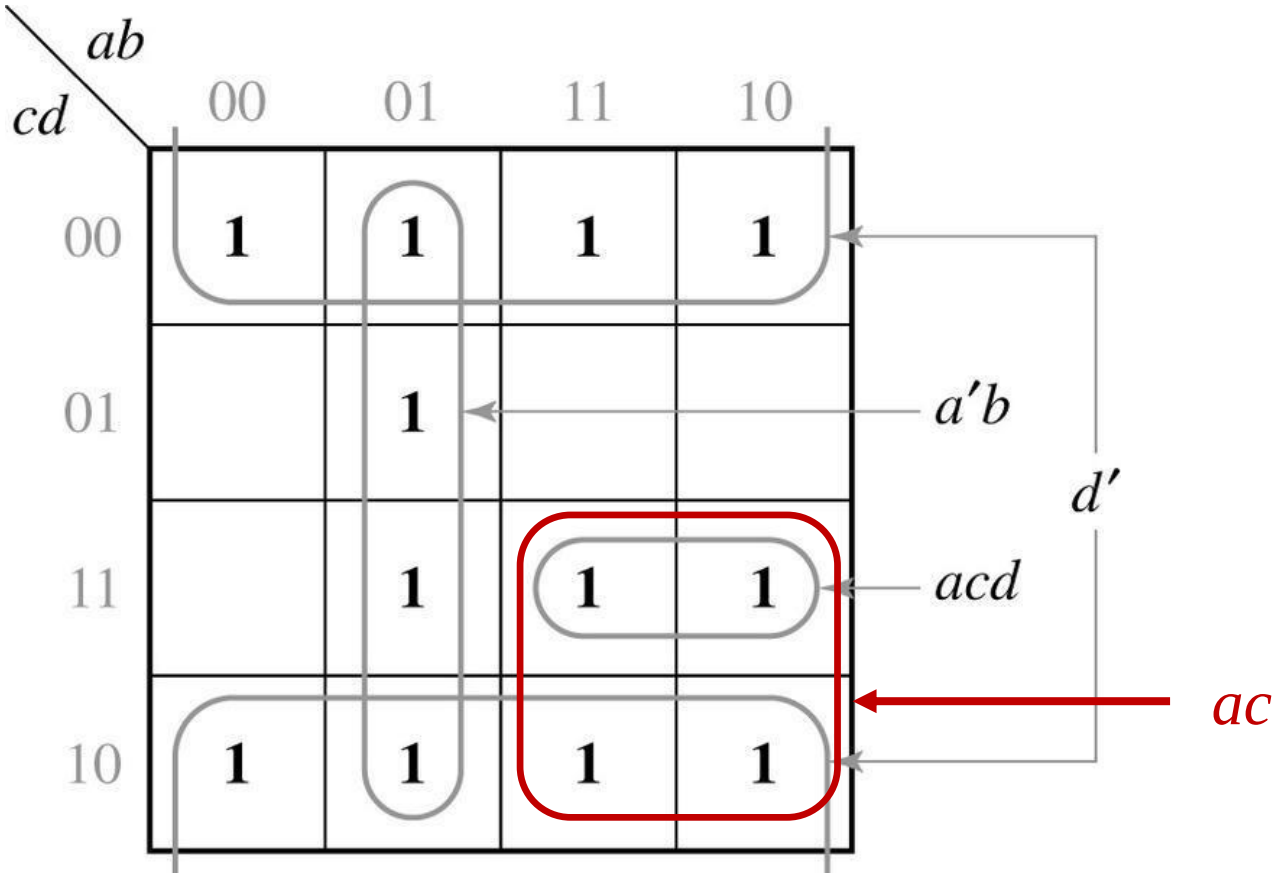
$$f(a,b,c,d) = acd + a'b + d'$$



Ex: 4-Variable Karnaugh Map (3)

Given expression of circuit, **modify** the expression using Karnaugh map

$$f(a,b,c,d) = acd + a'b + d' \quad \Rightarrow ac + a'b + d'$$



Don't Care Condition

Don't cares are optional in grouping.

$$f = \sum m(1,3,5,7,9) + \sum d(6,12,13) \quad \Rightarrow a'd + c'd$$

cd \ ab	00	01	11	10
00			X	
01	1	1	X	1
11	1	1		
10		X		

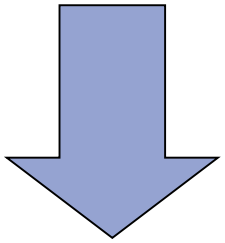
Including a don't care
= > *c'd*

Not including a don't care
= > *b'c'd*

Product of Sums Form

$$0\text{'s of } f$$

$$f' = y'z + wxz' + w'xy$$



complement

yz \ wx	00	01	11	10
00	1	1	0	1
01	0	0	0	0
11	1	0	1	1
10	1	0	0	1

$$f = (y + z')(w' + x' + z)(w + x' + y')$$

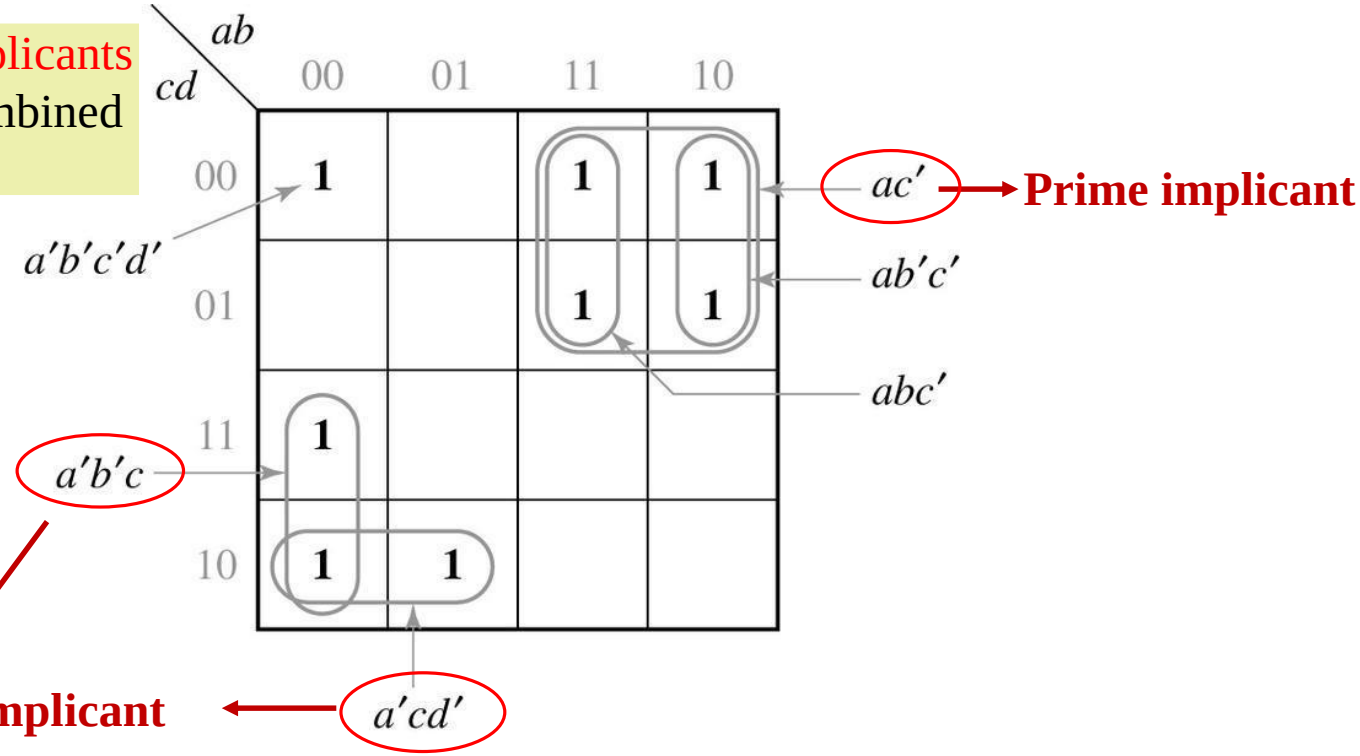
minimum product of sum for f

Prime Implicants

- **Implicants of F** : Any single '1' or any group of "1's which can be combined together on a Map

- **Prime Implicants of F** : A product term if it can not be combined with other terms to eliminate variable.

It is not Prime implicants since it can be combined with other terms

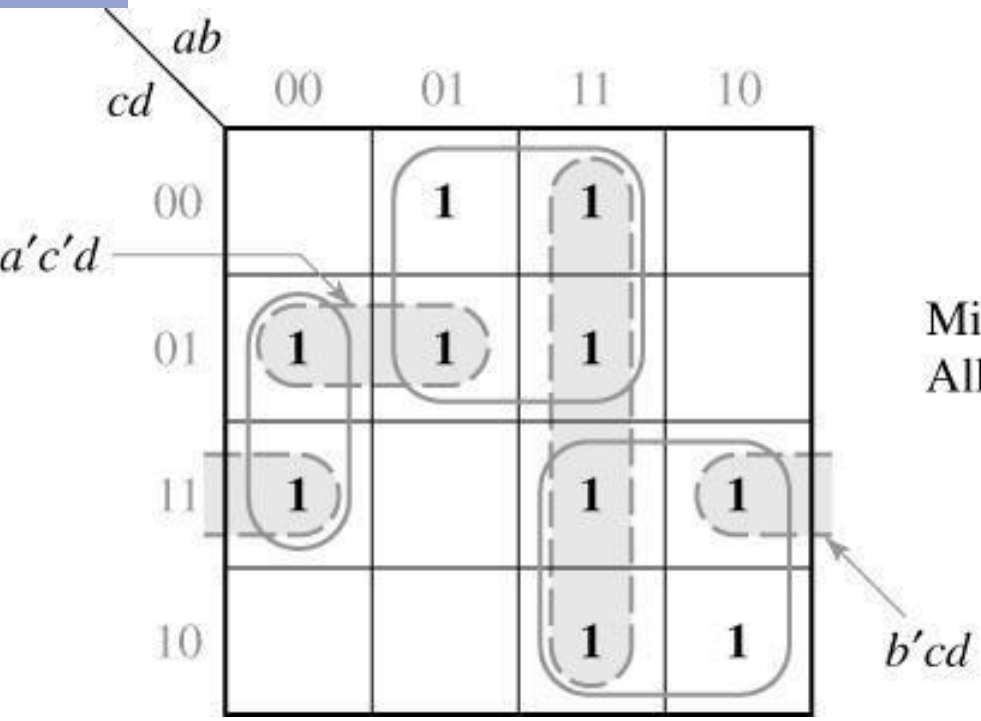


Minimum Sum of Products

The minimum sum-of-products expression consists of some (*not necessarily all*) of the prime implicants of a function.

– minimum number of prime implicants to cover all the 1's

If a minterm is covered by only one prime implicant, that prime implicant is *essential.*

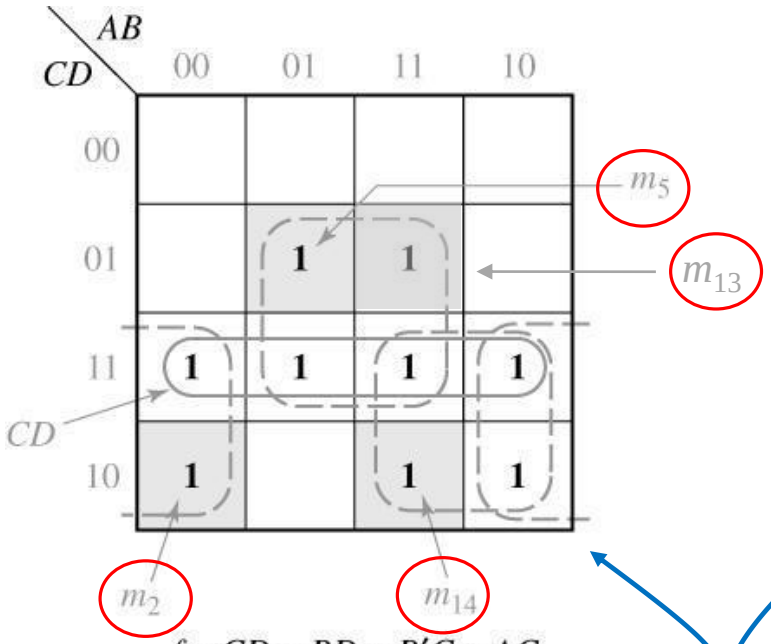


Minimum solution: $F = a'b'd + bc' + ac$

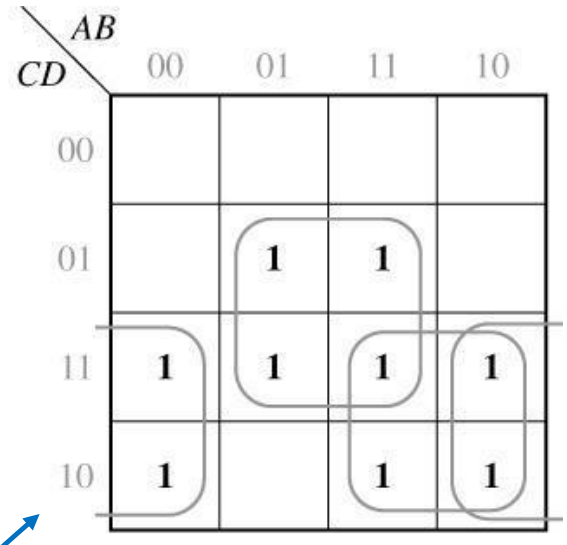
All prime implicants: $a'b'd$, bc', ac, $a'c'd$, ab, $b'cd$

Essential Prime Implicants

Because all of the prime implicants of a function are generally not needed in forming the minimum sum of products, selecting prime implicants is needed.



$f = CD + BD + B'C + AC$



$f = BD + B'C + AC$

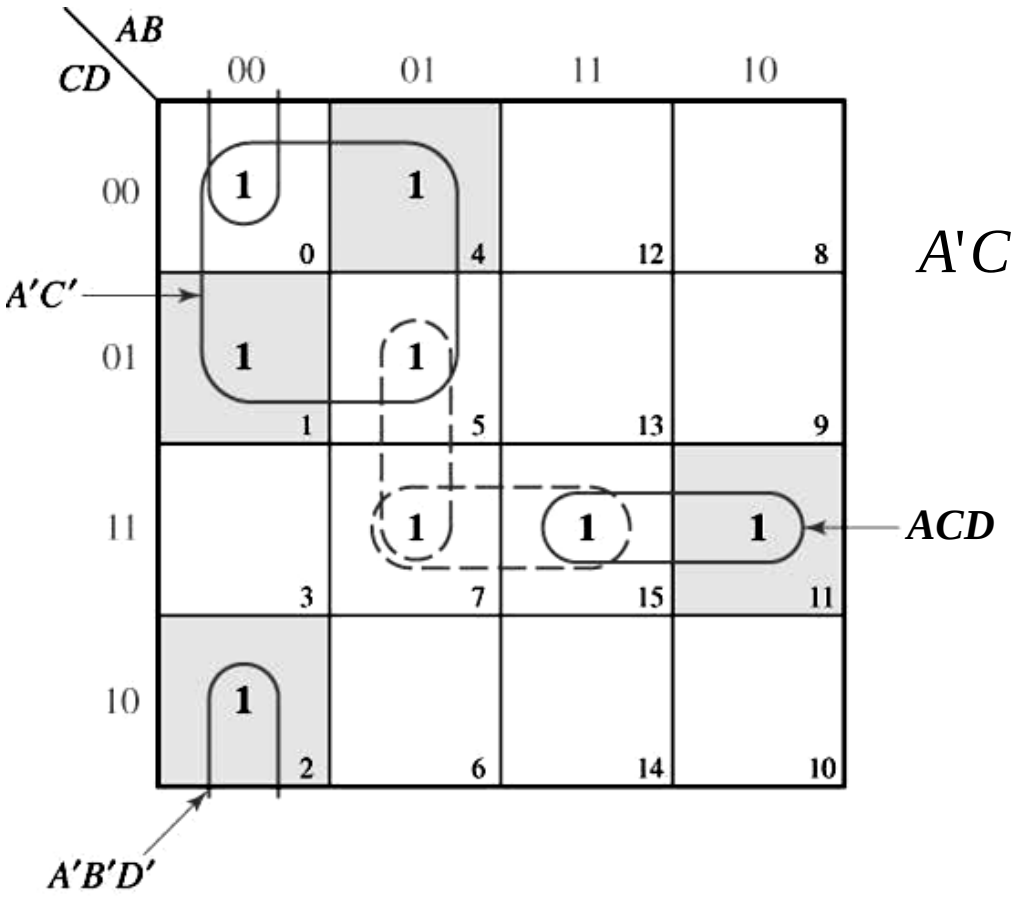
- ***CD*** **is not needed to cover for minimum expression**

-***B'C*** **,** ***AC*** **,** ***BD*** **are "essential" prime implicants**

- ***CD*** **is not an "essential" prime implicants**

General Procedure (Rule)

1. First, find essential prime implicants
2. If minterms are not covered by essential prime implicants only, more prime implicants must be added to form minimum expression.



$$A'C + A'B'D' + ACD + \begin{cases} A'BD \\ \text{or} \\ BCD \end{cases}$$

Don’t Care Condition

- Including don’t care terms to make larger groups
- Don’t care terms are not essential

Example

CD \ AB	00	01	11	10
00	X_0	1_4		1_8
01		1_5	1_{13}	1_9
11		X_7	X_{15}	
10		1_6		1_{10}

Essential: 1_6, 1_{10}

5-Variable Karnaugh Map

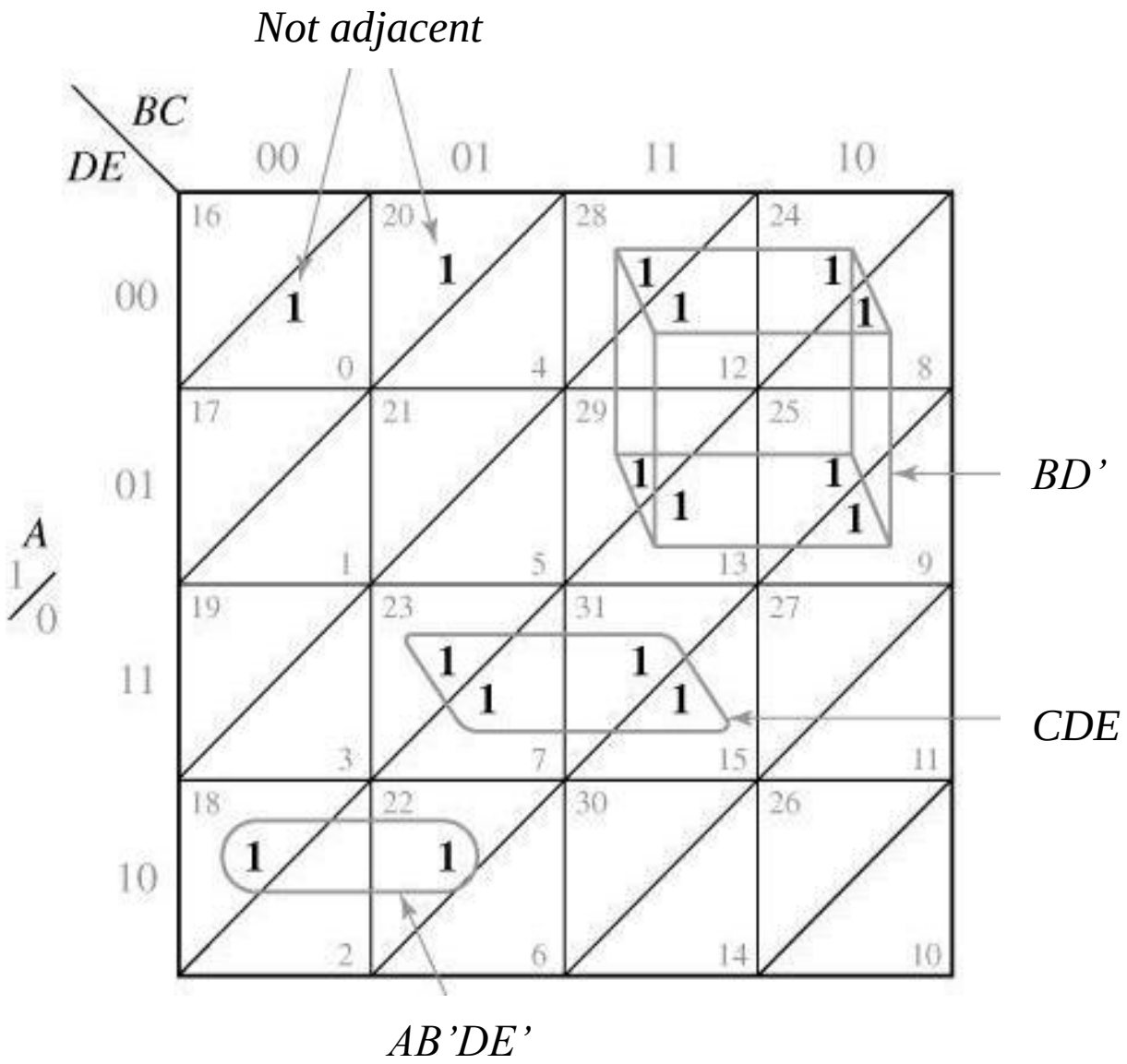
- Diagonal separation of each partition for 4 variables K. Map
- Two diagonal terms in a partition are adjacent.

ABCDE

A(1/0)

DE \ BC	00	01	11	10
00	16 / 0	20 / 4	28 / 12	24 / 8
01	17 / 1	21 / 5	29 / 13	25 / 9
11	19 / 3	23 / 7	31 / 15	27 / 11
10	18 / 2	22 / 6	30 / 14	26 / 10

Ex: 5-Variable Karnaugh Map (1)



Ex: 5-Variable Karnaugh Map (1)

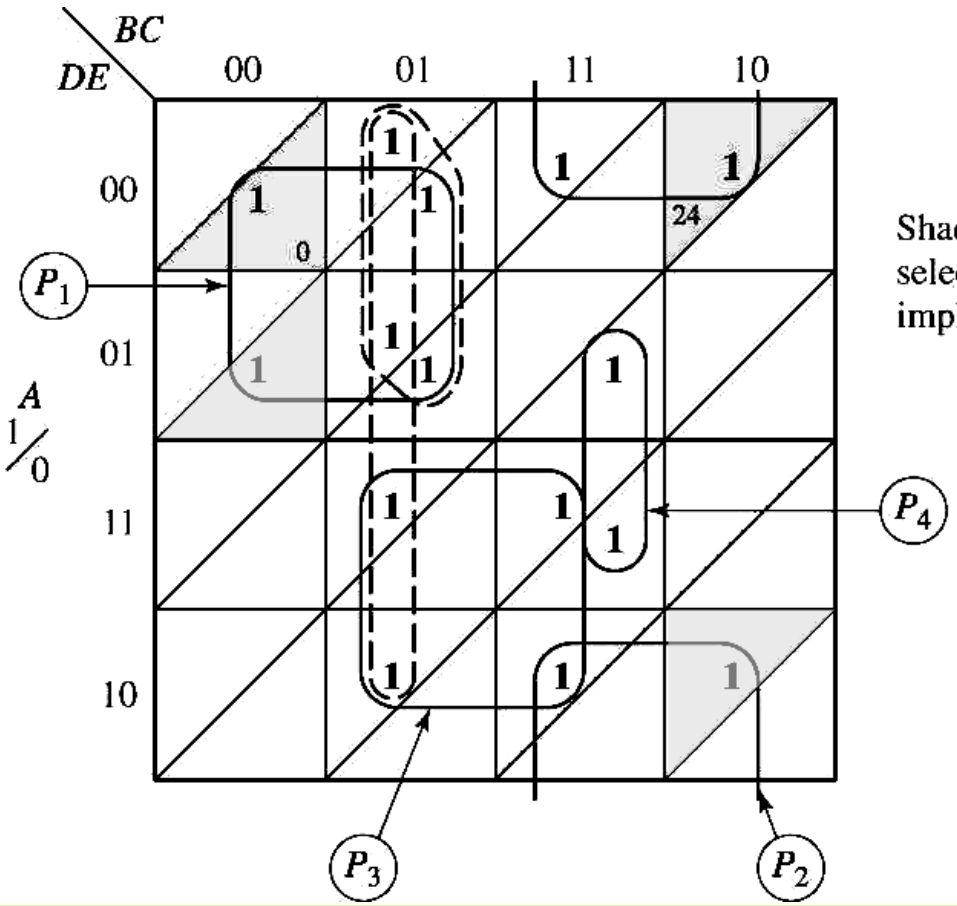
$$F(A,B,C,D,E) = \sum m(0,1,4,5,13,15,20,21,22,23,24,26,28,30,31)$$

ABCDE

A(1/0)

DE \ BC	00	01	11	10
00	1	1 1	1	1
01	1	1 1	1	
11		1	1 1	
10		1	1	1

Ex: 5-Variable Karnaugh map (2)



Shaded 1's are used to select essential prime implicants.

$$F = \underset{P_1}{A'B'D'} + \underset{P_2}{ABE'} + \underset{P_3}{ACD} + \underset{P_4}{A'BCE} + \begin{cases} AB'C \\ \text{or} \\ B'CD' \end{cases}$$

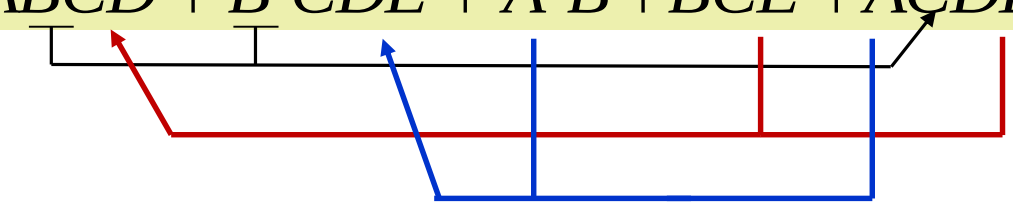
Other use of Karnaugh map (1)

Find the redundant terms in the expression.

$$F = ABCD + B'CDE + A'B' + BCE'$$

Using the consensus theorem:

$$F = ABCD + B'CDE + A'B' + BCE' + ACDE$$



by the consensus theorem

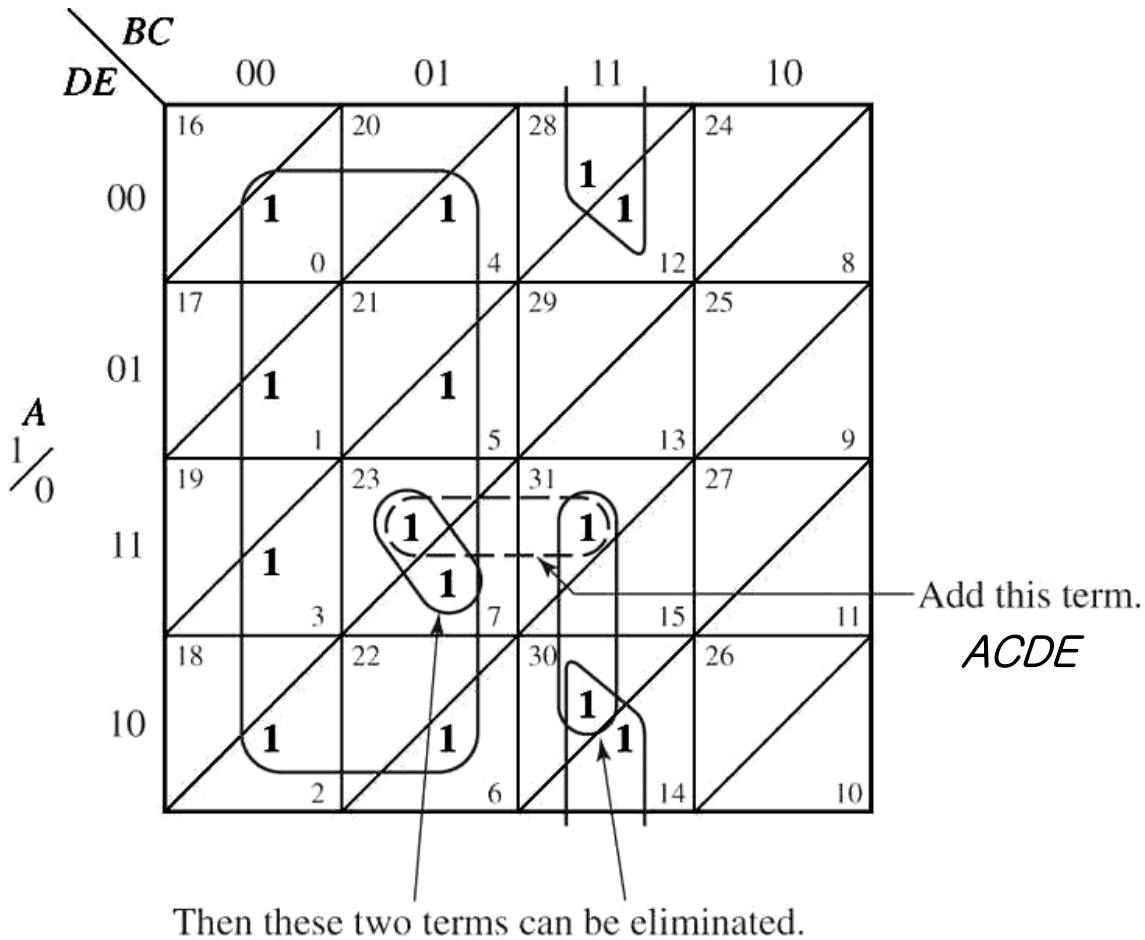
ABCD, B'CDE are eliminated

minimum solution:

$$F = A'B' + BCE' + ACDE$$

Other use of Karnaugh map (2)

Find the redundant terms in the expression.

$$F = ABCD + B'CDE + A'B' + BCE'$$

(30,31) **(7,23)** **(0-7)** **(12,14,28,30)**